We're constantly learning more about Space, and every new snippet of information challenges existing theories

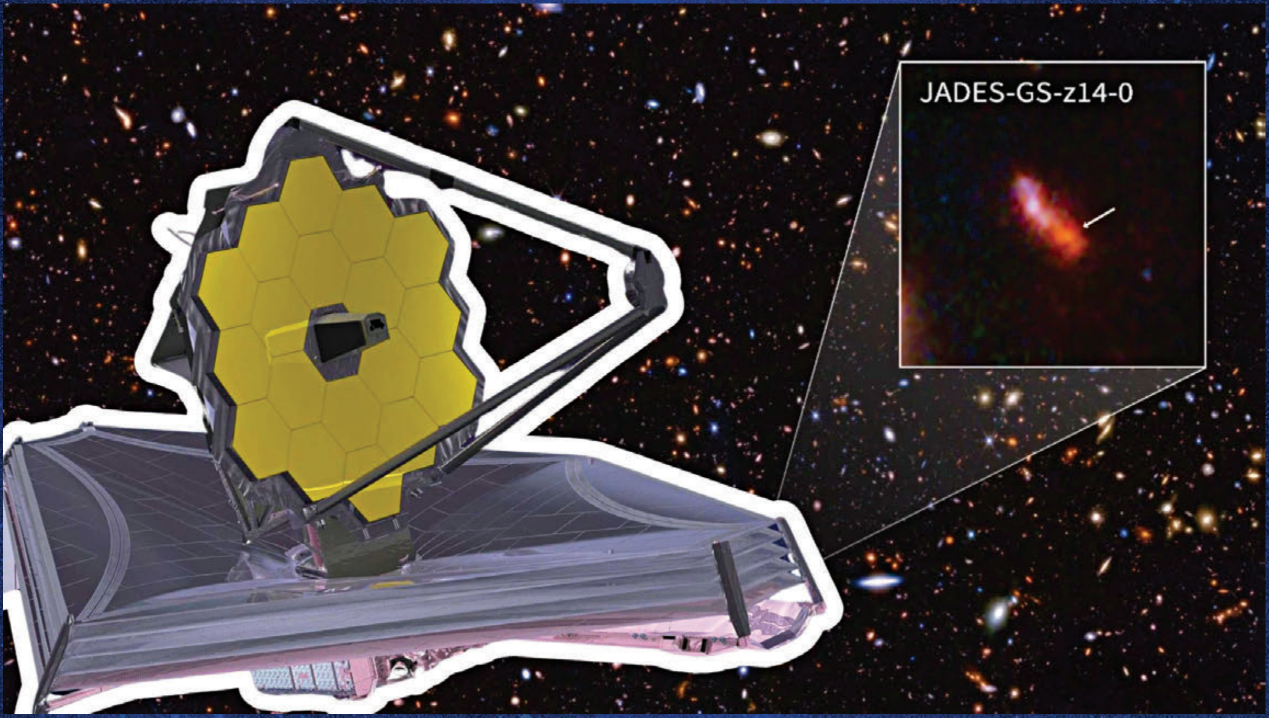

Image via University of Arizona

Oldest galaxy in universe

James Webb Telescope discovery to rewrite cosmic history?

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The cosmos, in its infinite complexity, has once again defied expectations. Astronomers using the James Webb Space Telescope (JWST) have uncovered something extraordinary: a primordial galaxy, JADES-GS-z14-0, existing less than 300 million years after the Big Bang. Not only does this discovery push the boundaries of our observational reach, but it also challenges established models of early galaxy formation and chemical evolution. The implications are profound, offering a rare glimpse into the infancy of the universe. These findings were published by a group of astronomers from the University of Arizona, in their research paper titled – Photometric detection at 7.7 μm of a galaxy beyond redshift 14 with JWST/MIRI - published in the journal Nature Astronomy.

The most distant known galaxy

To appreciate the magnitude of this find, one must first understand the significance of redshift. In cosmology, redshift (z) measures how much a galaxy's light has been stretched as the universe expands. Until recently, the observational frontier of galaxy formation was thought to extend to redshifts of around $z \approx 10$. However, the discovery of JADES-GS-z14-0 at $z = 14.32$ shatters this limit, placing it at a time when the universe was just a fraction of its current age. This makes it the most distant galaxy with a spectroscopically confirmed redshift.

A galaxy that shouldn't exist?

JADES-GS-z14-0 is not just remarkable for its distance – it is unexpectedly bright and chemically evolved for such an early cosmic epoch. Typically, galaxies forming shortly after the Big Bang are expected to be small, dim, and composed mostly of hydrogen and helium, the first elements forged in the universe. Yet, JWST's Mid-Infrared Instrument (MIRI) has